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Simulation and selection of static mixer, the core equipment of middle-low temperature coal tar pretreatment, based on the computational fluid dynamics

Yijie Wang [a,b], Dong Li [a,b,*], Yonghong Zhu [a,b,*], Jieping Liu [a,b], Hongyu Chen [c], Qing Guo [a,b], Chongpeng Du [a,b], Zhengpeng Miao [a,b], Louwei Cui [a,b], Feng Tian [a,b], Jie Liu [a,b], Huaan Zheng [a,b]

[a] School of Chemical Engineering, Northwest University, Xi'an 710069, Shaanxi, China
[b] The Research Center of Chemical Engineering Applying Technology for Resource of Shaanxi, Xi'an 710069, Shaanxi, China
[c] School of Aerospace Engineering Xiamen University, Xiamen 361102, Fujian, China

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ABSTRACT

The static mixer is first introduced into the middle-low temperature coal tar acid refining pretreatment process to solve the problem of uneven mixing of high viscosity coal tar and additives. The mixing process of coal tar and additives in four static mixers was simulated by the CFD software. Through the volume fraction cloud diagram of coal tar phase, it's found that the mixing effect of SX static mixer is better. For SX static mixer, the axial positions where pure additives and pure MTCT no longer exist are 96.3mm and 190.7mm respectively. Then quantitatively characterize outlet cross section of static mixer, the results show that the separation strength of SX static mixer is the lowest, which is 3.5×10^{-5}. After comprehensive research, it's finally determined that when the number of SX mixing elements is three and installed at 90° without gap, the comprehensive performance is better. A pilot experimental device for coal tar pretreatment is designed and built. Taking the removal effect of Fe heteroatoms after coal tar pretreatment as the detection index for simulation verification. The experimental results show that the removal rate of Fe heteroatoms in the pilot plant with SX static mixer is the highest.

1. Introduction

In recent years, domestic dependence on foreign oil has increased year by year, reaching a record high of 73.5% in 2020 [1]. The contradiction between oil supply and demand is still one of the key factors restricting China's economic development. China is rich in coal resources, of which low rank coal accounts for more than half, mostly distributed in Northwest and Northeast of China, especially in Yuling Shaanxi, Ordos Inner Mongolia and Xinjiang. Low rank coal can be converted into high value-added products such as gas, tar and semi-coke by medium and low-temperature distillation technology, to realize effective utilization of quality and classification. The Action Plan for Energy Technology Revolution and Innovation (2016-2030) organized by the National Development and Reform Commission and the National Energy Bureau lists the classification and effective utilization of low rank coal as a key research and development-key research-demonstration technology. The preparation of clean fuel oil from coal tar can alleviate the oil shortage and fill the energy shortage in China to a certain extent, which is of great significance and value for sustainable development.

The middle-low temperature coal tar (MTCT) is a by-product of coal pyrolysis and coking, with an annual output of more than 10 million tons in China [2]. There are many metal heteroatoms such as Fe and Ca and nonmetal heteroatoms such as S and N in MTCT, which are difficult to remove, especially oil-soluble metals, basic nitrogen compounds and thiophene-type sulfide, which greatly limits the subsequent processing of coal tar into high value-added products [3–5]. Many scholars have developed a series of coal tar pretreatment processes to remove heteroatoms [6–9]. Among them, the treatment method of acid refining method is simple and convenient [10–11], and the improved acid refining method developed by Liu [12] showed an superior removal effect.

The mixing of coal tar and additives is an important process in acid

Abbreviations: MTCT, Middle-low temperature coal tar; SM, Static mixer; SVSM, SV static mixer; SXSM, SX static mixer; SLSM, SL static mixer; SKSM, SK static mixer; CFD, Computational fluid dynamics.

* Corresponding author.
E-mail addresses: lidong@nwu.edu.cn (D. Li), 1070981251@qq.com (Y. Zhu).

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Nomenclature

$C_{1\varepsilon}$	turbulence model constant
$C_{2\varepsilon}$	turbulence model constant
C_m	turbulence model constant
D	mixer diameter
G	turbulent kinetic energy generated due to mean velocity gradient
J_i	diffusive flux
J_i	diffusive flux
P	pressure
Re	Reynolds numbers
S_i	creation rate from the dispersed phase plus the user-defined sources
t	time
u	velocity
Y_i	mass fraction of the species i

Greek symbols

ε	turbulent kinetic energy dissipation rate
σ_k	turbulent Prandtl number of k
σ_ε	turbulent Prandtl number of ε
μ	viscosity
μ_t	turbulent viscosity
ρ	fluid density

refining pretreatment process. The uniformity and stability of mixing directly affect the removal effect of heteroatoms, and affect the subsequent deep processing and utilization of coal tar. Compared with the classical dynamic mixer, the static mixer (SM) has the advantages of no moving components, low maintenance cost, low energy consumption and low space cost [13–14]. Many scholars [15–17] have studied the principle of SM. Through its built-in mixing element, the fluids flow state in the pipe is changed to realize the mixing or reaction of various fluids, including domestic SVSM, SXSM, SLSM and SKSM can be used for liquid-liquid mixing [18]. Due to the unique advantages of SM, it has a large number of applications in industrial production.

If computational fluid dynamics (CFD) software is used to combine visualization and numerical calculation to simulate SM, the loss of manpower and material resources can be reduced [19–21]. Jones et al. [22] focused on coagulation and disinfection in wastewater and water treatment, established a CFD model to predict the internal turbulence of the dual spiral SM. The results show that the mixing intensity of the SM largely depends on the selection of the mixer volume. In order to optimize the performance of real-time mixing sprayer in orchard, Li et al. [23] used SIMPLEC algorithms to simulate the pesticide and water in SKSM, SXSM and SDSM. It was concluded that the real-time mixing performance of SX static mixer was better, and the optimal number of units was 5. The CFD model was verified by comparing the simulation results with the concentration data of tracer in mixed drug test. F. Theron et al. [24] studied the pressure drop and emulsification performance of cyclohexane and water in turbulent state in three different SMs: SMX, SMV and SMXPlus. It was found that the pressure drop of SMX SM is the highest. When the concentration of dispersed phase is 10% to 60%, the droplet distribution does not show coalescence. Scholars have obtained many valuable conclusions by using CFD software to study the SM, which greatly popularized the application of the SM. However, people mainly simulate the mixing process of crude oil, wastewater, drugs and other fluids in the SM, CFD software is rarely used to simulate the mixing process of low-density mixed fluids such as coal tar and additives in SM. Giuseppina et al. [25] studied the mixing of two liquids with different densities and viscosities in the pipe of SMV SM by using CFD simulation based on Reynolds Averaged Navier Stokes. The results show that the density difference may lead to significant changes in the mixing degree, and also proposed an improved mixing degree evaluation method. This study is carried out on a simple geometry containing an SMV component, which can be easily extended to the combination of other mixing elements. At the same time, it also has some limitations. It does not investigate the effect of variables such as the type and number of mixing elements on the performance improvement of SM.

On the basis of previous work, the CFD software was used in this study to simulate and analyze the mixing effect of coal tar and additives in SVSM, SXSM, SLSM and SKSM with single component, and select the best SM for mixing. Because the design and structural optimization of the SM can reduce energy consumption and achieve higher mixing performance, the flow field analysis and structural optimization of the SM with the best mixing effect are studied under the same number of components. The influence of different number of mixing elements in SM on the mixing effect, pressure loss and turbulent kinetic energy of coal tar and additives are analyzed, and the influence of geometric layout such as installation spacing of mixing elements on the mixing effect are analyzed. Finally, the verification experiment is carried out based on the simulation results. In order to deepen the understanding of the mixing mechanism of coal tar and additives, optimize the process conditions and improve the pretreatment device, provide a theoretical basis for the research of coal tar pretreatment process, and provide reference for the selection of SM in the mixing process of other fluids with high viscosity and low-density difference in industrial production.

2. Geometric conditions and meshing

2.1. Physical model

When using SM for material mixing, there are usually two mixing methods. One is that heavy oil and dilute oil are mixed before entering the SM [26], and the other is mixed in the SM. The mixing mode will directly affect the mixing efficiency of the SM. Wang et al. [27] pointed out that when two streams of fluids are mixed, the mixing effect was the best when the material was added at the pipe wall position. Therefore, this paper adopts the in pipe mixing mode of two material inlets.

According to the actual size of the mixer in the pilot plant for coal tar pretreatment, the geometric models of four mixing elements are established, as shown in Fig. 1. Then assemble four SV, SX, SL or SK mixing elements into the main pipe, respectively, and the three-dimensional model of the SM is shown in Fig. 2. Coal tar and additives are mixed at 80mm away from the inlet of the main pipe. The four internal components of the SM are also arranged and assembled at 80mm away from the inlet of the main pipe.

2.2. Meshing

In this paper, the geometric model of the SM is divided by unstructured grid. As shown in Fig. 3, seven models with different grid sizes are established for SXSM, and the grid independence test is carried out with the mixing effect (separation strength) when the fluids flow through the last mixing element as an index. It can be seen that with the increase of the number of grids, the separation strength gradually tends to be stable. When the number of grids is greater than or equal to 1.5 million, the change of separation strength is very small, indicating that the calculation accuracy is not affected by the number of grids at this time. In addition, we also tested the grid independence of SVSM, SLSM and SKSM according to the above similar steps. It is found that when the number of grids is greater than 1.5 million, the separation strength of the four static mixers is gradually stable, and the calculation accuracy is not affected by the number of grids.

Under the condition of ensuring the convergence speed and calculation accuracy, all the simulations in this paper control the number of grids to more than 1.5 million. The meshing results of the four SMs are shown in Fig. 4. The skewness distribution of different types of SMs is

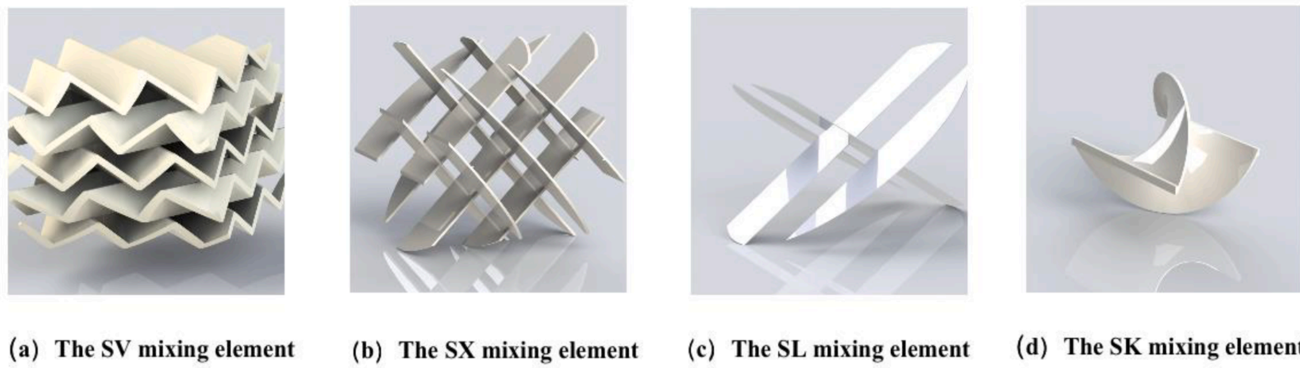

Fig. 1. SV, SX, SL or SK mixing element models (The thickness and height of the single corrugated plate of the SV-type hybrid element are 1.4 mm and 5 mm, respectively. The angle between the broken line of the corrugated plate and the horizontal plane is 45°, and the clearance rate is 0.909. The structure of SX type mixing element is formed by 90° opposite welding of baffle element plates and the baffle thickness is 2 mm. The structure of SL type mixing element is formed by 60° opposite welding of baffle elements, and the thickness of baffle is the same as that of SX-type hybrid element. The baffle thickness of SK type hybrid element is 1mm.).

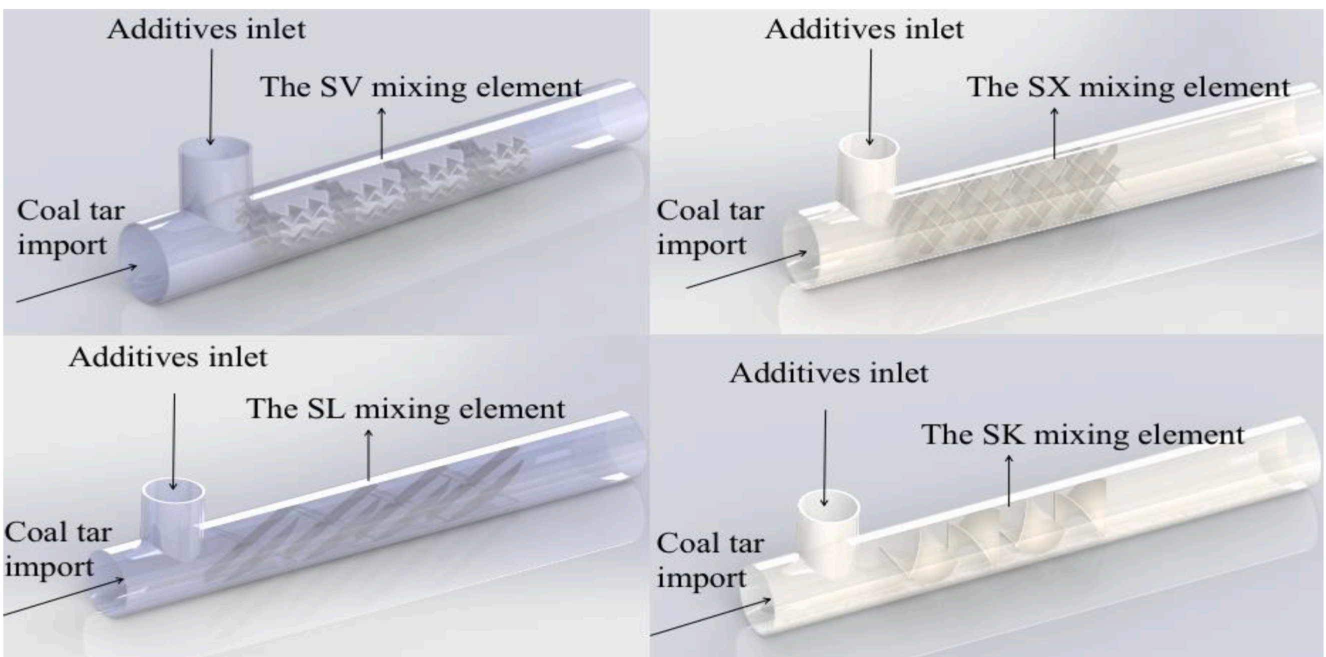

Fig. 2. SVSM, SXSM, SLSM and SKSM models (The main pipe is 400mm long, the branch pipe is 30mm long and the pipe diameter is 44mm. Each mixing element is arranged in 90° staggered arrangement without gaps.).

shown in Table 1. More than 99% of the mesh skewness after structure division is concentrated between 0 and 0.6, and the maximum skewness is less than 0.89, which means that the quality of the mesh is "good" and can be used for multiphase flow calculation [28].

3. Simulation program

3.1. Mathematical model

Before using the CFD software for simulation, the mathematical model should be established according to the experimental conditions based on the basic principles of fluid mechanics. This paper studies the mixing of high viscosity coal tar and additives in SM, and the temperature change was not considered. Based on the CFD simulation of Meng et al [25,29-31], in order to reduce the calculation complexity and save the computational capacity the following assumptions are made for the calculation fluid:

(1) Coal tar and additives are incompressible fluids;
(2) The cross-sectional velocity distribution of the fluid inlet pipe is uniform;
(3) The fluid composition is uniform;
(4) The fluid flows continuously;

In the present work, the CFD software is applied to analysis the mixing of high viscosity coal tar and additives in SM. The mass, momentum, energy and component conservation equations and the equations of Reynolds numbers Re and turbulence model are written as follows [24,30-31]:

Continuity equations:

$$\frac{\partial \rho}{\partial t}+\frac{\partial \rho u_i}{\partial x_i}=0 \tag{1}$$

Momentum conservation equation:

$$\frac{\partial(\rho u_i)}{\partial t}+\frac{\partial(\rho u_i u_j)}{\partial x_i}=-\frac{\partial P}{\partial x_i}+\frac{\partial}{\partial x_i}\left[(\mu+\mu_t)\left(\frac{\partial u_i}{\partial x_i}+\frac{\partial u_j}{\partial x_i}\right)\right] \tag{2}$$

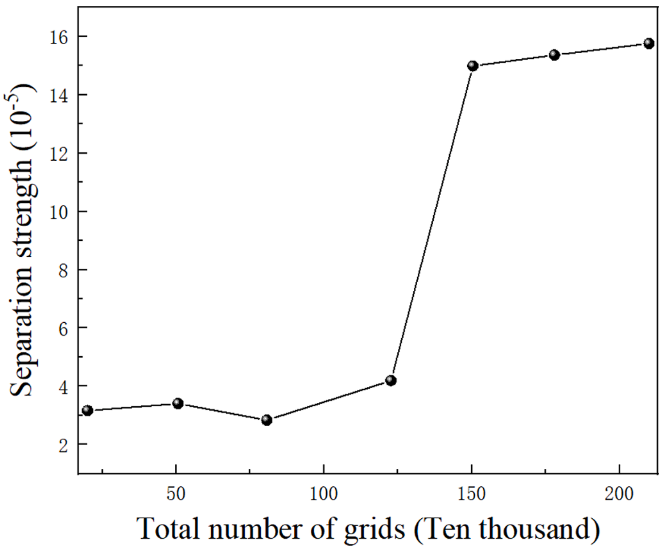

Fig. 3. Variation of separation strength with mesh density.

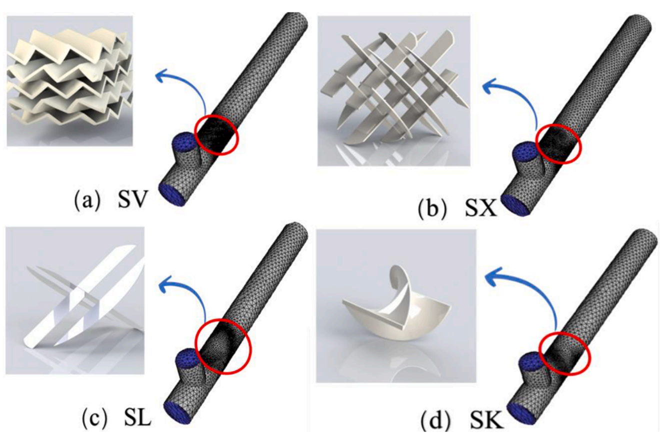

Fig. 4. Meshing of SVSM, SXSM, SLSM and SKSM.

Table 1
The grid skewness distribution of different mixer structures.

Skewness	0~0.2	0.2~0.4	0.4~0.6	0.6~0.8	0.8~0.1
SK	36.03%	44.80%	18.72%	0.45%	0.00%
SV	35.90%	56.87%	6.85%	0.38%	0.00%
SL	42.19%	45.82%	11.48%	0.51%	0.00%
SX	45.27%	45.20%	9.18%	0.35%	0.00%

The mass transport equations:

$$\frac{\partial(\rho \mathrm{Y_i})}{\partial t} + \nabla\cdot(\rho \mathrm{V} \mathrm{Y_i}) = -\nabla\cdot \mathrm{J_i} + \mathrm{R_i} + \mathrm{S_i} \tag{3}$$

The Reynolds numbers Re:

$$\mathrm{R}_e = \frac{\rho u D}{\mu} \tag{8}$$

The transport equation for the turbulent kinetic energy (k) is as follows:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_i k)}{\partial x_i} = \frac{\partial}{\partial x_i}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_i}\right] + \mathrm{G} + \rho\varepsilon \tag{4}$$

The transport equations for the turbulent dissipation rate (ε) are as follows:

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \frac{\partial(\rho u_i \varepsilon)}{\partial x_i} = \frac{\partial}{\partial x_i}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_i}\right] + C_{1\varepsilon}\cdot\frac{\varepsilon}{k}\mathrm{G} - C_{2\varepsilon}\cdot\frac{\varepsilon^2}{k}\rho \tag{5}$$

The viscosity coefficient equation:

$$\mu_t = \rho C_m \frac{\mathrm{k}^2}{\varepsilon} \tag{6}$$

The Turbulent kinetic energy caused by average velocity gradient:

$$G = \mu_t\left(\frac{\partial \mu_j}{\partial x_i} + \frac{\partial \mu_i}{\partial x_i}\right)\frac{\partial \mu_i}{\partial x_j} \tag{7}$$

Where P, ρ, t, μ and D are respectively the pressure, fluid density, time, viscosity coefficient and the mixer diameter; μ_t is turbulent viscosity; Y_i is local mass fraction of each phase, J_i is the diffusive flux, R_i is the rate produced by chemical reaction, S_i is the creation rate from the dispersed phase plus the user-defined sources.; u_i *is the* velocity components in the triple Cartesian directions, i.e., x, y, and z directions; G is the turbulent kinetic energy generated due to mean velocity gradient; The model constants are $C_m = 0.09$;$\sigma_k = 1.0$;$\sigma_\varepsilon = 1.3$;$C_{1\varepsilon} = 1.44$ and $C_{2\varepsilon} = 1.92$.

3.2. Boundary condition

Considering that coal tar and additives have certain density difference and velocity difference, the mixture model [32] in multiphase flow model is selected for numerical calculation of velocity field and concentration field of MTCT-additives two-phase flow. According to the experimental conditions, the two inlets are set as the velocity inlet. The coal tar is set as the main phase, inlet speed is $0.245\mathrm{m\bullet s^{-1}}$ and its Reynolds number is 531.The additives is the second phase, inlet speed is $0.069\mathrm{m\bullet s^{-1}}$, and its Reynolds number is 303. The volume fraction of the second phase at the coal tar inlet is 0, and the volume fraction of the second phase at the additive's inlet is set to 1. The pipe outlet is set as static pressure outlet and the value of the pressure outlet is a standard atmospheric pressure. The wall boundary condition is set as the WALL boundary without slip. MTCT-additives surface coefficient is 0.018N/m, oil-gas surface coefficient is 35N/m, and gas-water surface coefficient is 0.0728 N/m.

3.3. Numerical calculation method

The three-dimensional double precision separation solver was selected in the CFD software. The pressure and velocity fields were calculated by SIMPLE algorithm under incompressible, isothermal, and transient state conditions. And the volume fraction, momentum, turbulent kinetic energy, turbulent energy dissipation rate and Reynolds stress were discretized in the form of first order upwind. The turbulence model adopts the Realizable model, and the relaxation factor and convergence residual adopt the recommended values in the CFD software for simulation calculation. The physical parameters of the mixture are shown in Table 2.

4. Experimental verification

4.1. Experimental purpose

The coal tar pretreatment process unit is composed of feed part,

Table 2
Physical parameters of mixtures.

Name	Inlet speed $/\mathrm{m\bullet s^{-1}}$	Density $/\mathrm{kg\bullet m^{-3}}$	Viscosity /Pa•s	Inlet temperature /°C
Additives	0.069	1000	0.001	25
coal tar	0.245	1040	0.0211	80

mixing part and separation part. The mixing part is the stage of removing heteroatoms, and the separation part is the stage of separating MTCT-additives phase. Since the separator is unique, the separation effect remains unchanged. These two parts are particularly important, which are related to the effect of removing heteroatoms in coal tar pretreatment. It is not easy to accurately observe the mixing effect of static mixers installed in pilot plant. We decided to use the Fe heteroatoms removal effect after coal tar pretreatment to indirectly judge the mixing effect of static mixer.

4.2. Experimental device and method

Based on the physical model, a pilot plant is respectively equipped with SVSM, SXSM, SLSM and SKSM with one mixing element and equipped with SXSMs with two, three and four mixing elements for verification. The pilot plant setting is shown in Fig. 5 and the list of main devices is shown in Table 3. The process flow chart is shown in Fig. 6. The specific process flow is as follows: (1) In the feed part, the coal tar in the coal tar storage tank is preheated by the heating rod to the set temperature, and the additives that are uniformly stirred from the auxiliary tank are pumped into the static mixer by the feed pump with a certain mass ratio of coal tar and additives. (2) In the mixing part, there is no gap between the mixing elements. Under the installation condition of 90°, the two materials are fully mixed in the SM at the reaction temperature and flow rate provided in Table 2 to fully complete the heteroatom removal reaction. (3) In the separation part, the mixture after reaction enters the three-phase separator to complete the separation of heavy oil water light oil at the set temperature. The heavy oil phase is discharged from the sewage outlet, the water phase is discharged from the drain outlet, and the light oil phase is discharged from the coal tar oil outlet for sampling and analysis respectively.

The KY-3000 SN sulfur nitrogen analyzer of Jiangsu Keyuan Electronic Instrument Company was used to detect the sulfur content in MTCT. The specific operation steps are as follows: a certain amount of MTCT samples before and after pretreatment were placed in the reagent tube, and an appropriate amount of toluene solution was added for full dissolution and dilution to be analyzed and detected. Secondly, appropriate amount of samples to be analyzed were taken with a micro syringe and sent to the injection area for determination and analysis. and then the removal effect of Fe heteroatoms after pretreatment was calculated to judge the quality of coal tar acid refining pretreatment.

Fig. 5. Coal tar pretreatment unit.

Table 3
List of main devices.

Name	Specification and type	Quantity	Producer
Raw material tank	60L	2	Xi'an Hengxu Technology company
Auxiliary tank	30L	1	
Three phase separator	60L	1	
Diaphragm metering pump	JXM-A 5/1.0-PHS-O	1	Zhejiang ailipu Pump Industry company
Plunger metering pump	J1.6A 25/1.3-SHS- B	1	
Interface detector	BSPT 3/4	1	Beijing Hualan Innovation Technology company

5. Results and discussion

5.1. Influence of different mixers on mixing effect

CFD simulation of SVSM, SXSM, SLSM and SKSM was carried out to investigate the flow characteristics and mixing effect of coal tar and additives.

5.2. The mixing effect of coal tar and additives in SVSM

The volume fraction of coal tar phase in SVSM is shown in Fig. 7. Coal tar and additives flow into the SM from the main pipe and branch pipe respectively. Before entering the mixing element, the two different fluids basically do not mix and the interface starts at the junction of the branch pipe and the main pipe, and gradually transitions downward from the upper pipe wall. It can be seen from Fig. 7 (a) that after the liquids enter the first mixing element, coal tar and additives are mixed. This is because the SV mixing element is composed of 45° inclined corrugated plate, which forces the flow direction of materials to change. Under the action of stacking and twisting of corrugated sheets, coal tar and additives flow forward in the trend of mutual diffusion and mixing. At the beginning, only a small amount of additives and coal tar were mixed, concentrated in the middle and upper part of the pipeline. As the mixture flows forward the flow equilibrium, the mixing effect closer to the outlet gradually becomes better. Additives in pure form no longer exist at a distance beyond 146.6 mm at the entrance of the main pipe. It can be seen from the outlet section that there is a small amount of pure coal tar at the bottom of the pipeline, and additives and coal tar are poorly mixed. It can be seen from Fig. 7 (b) that when two SV mixing element were staggered, the material fluctuates in direction of fluids flow and superimposes transverse fluctuations. From the volume fraction of coal tar phase in the outlet section, it can be seen that the volume fraction of coal tar phase in the SV mixer is up to 94.74 %, and there are fewer layers in the cloud diagram. Adding two SV mixing element, the volume fraction of coal tar phase is closer at 0.842, and the mixing effect is significantly better than that of single component.

5.2. The mixing effect of coal tar and additives in SXSM

Fig. 8 shows the volume fraction of coal tar phase flowing through SX mixing element. The distribution ranged from 78.95% to 84.21%. There is a fluid boundary below the junction between branch pipe and main pipe, indicating that coal tar and additives have not been mixed. When the fluids flow through SX mixing element arranged in the pipe, the two-phase flow begins to mix. When the fluids pass through a single SX mixing element, there is a strong mixing effect, and additives in pure form no longer exist at a distance beyond 96.3 mm at the entrance of the main pipe. When the fluids flow through about 190.7 mm, the pure MTCT phase begins to disappear. After the fluids passes through two staggered SX mixing elements, the uneven mixing of additives at the top and coal tar at the bottom of the pipeline disappears. The mixing of

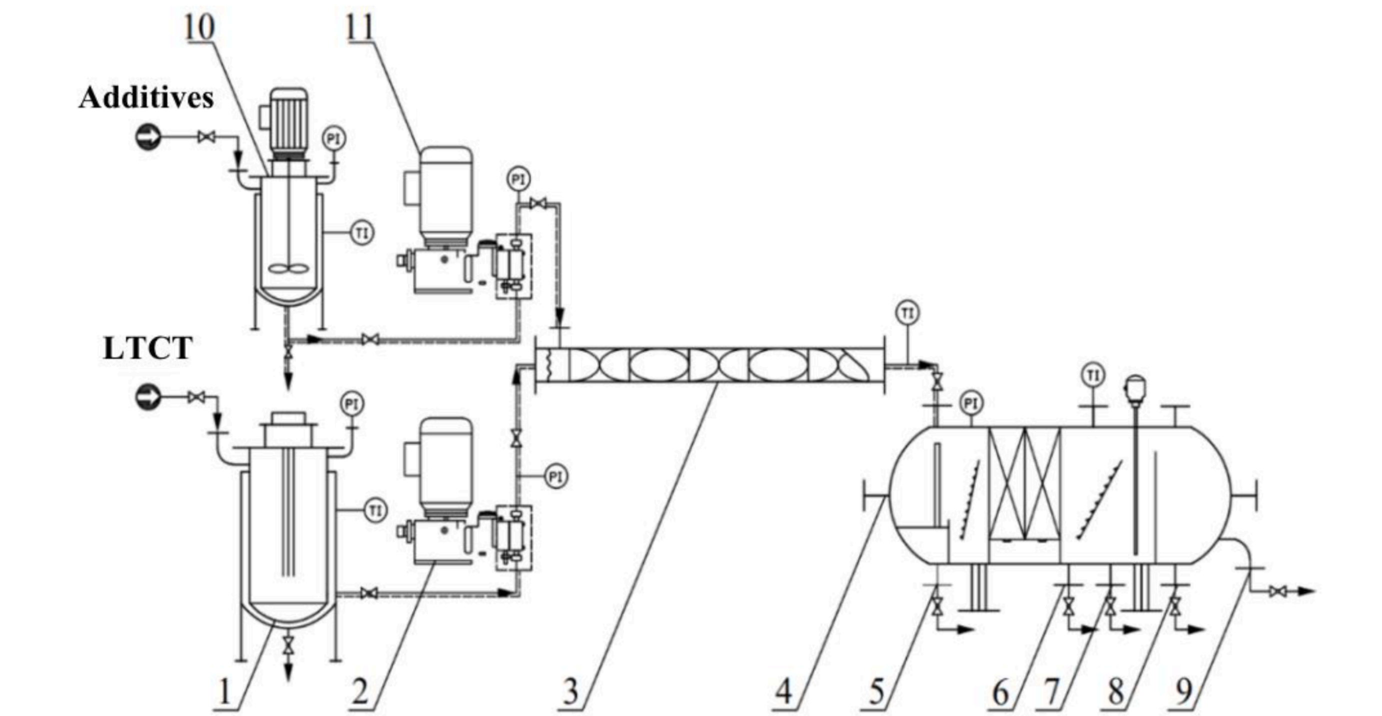

Fig. 6. Process flow chart of coal tar pretreatment (1—Coal tar storage tank, 2—Coal tar pump, 3—SM, 4—High efficiency oil-water separator, 5—Sewage outlet 1, 6—Sewage outlet 2, 7—Drain outlet, 8—Sewage outlet 3, 9—Coal tar outlet, 10—Auxiliary tank, 11—Auxiliary pump).

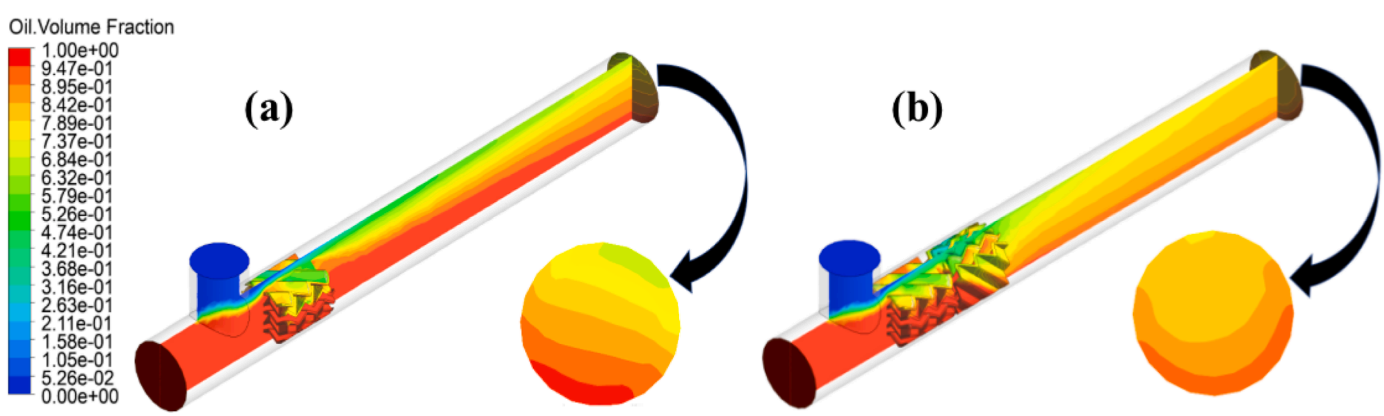

Fig. 7. Volume fraction of coal tar phase in SVSM (a for one SV mixing element, b for two SV mixing elements).

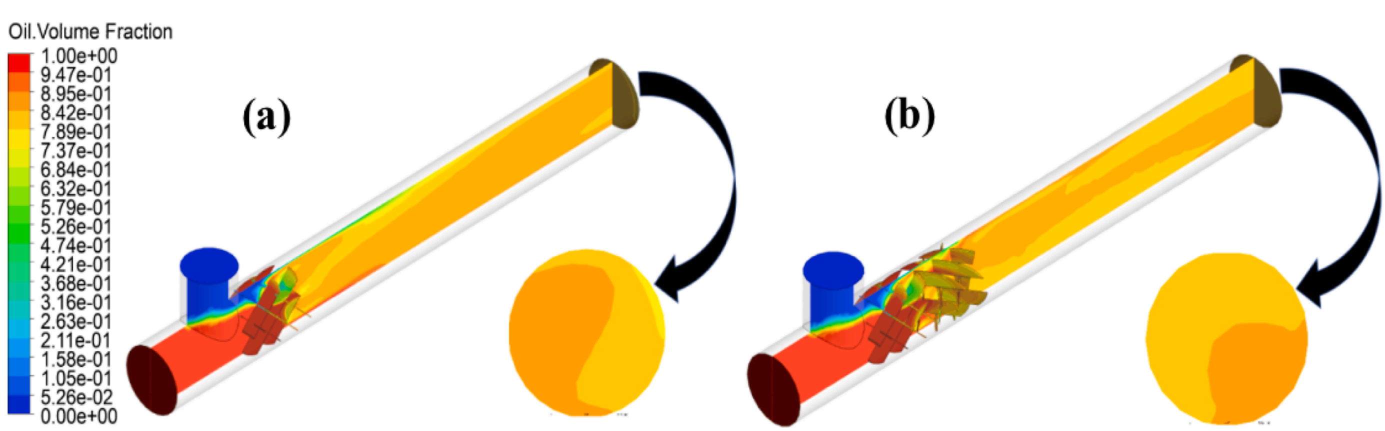

Fig. 8. Volume fraction of coal tar phase in SXSM (a for one SX mixing element, b for two SX mixing elements).

materials can be more sufficient and stable with a rapid trend. From the outlet section, the mixing effect of SXSM is better than that of SVSM, because the SX mixing element can guide the material flow to mix staggered from two dimensions.

5.3. The mixing effect of coal tar and additives in SLSM

Fig. 9 shows that the volume fraction of coal tar in SLSM is in the same range as that in SXSM. Compared with other types of SMs, the additives are not greatly disturbed when entering the SL mixing elements. They mainly remain on the upper channel of the SL component to restrict the forward flow of the additives, and then mix the additives with the coal tar at the bottom of the component. When the fluids flow through about 113.8 mm, the additives in pure form begin to disappear and when fluids flow through 207.1 mm, the pure MTCT also begin to disappear. The coal tar after mixing is mainly stabilized within the expansion width of the parallel baffle of the SL component, and the uniformly mixed phase gradually diffuses to the edge of the pipe. In general, the mixing effect of SLSM is good and can tend to stabilize quickly.

5.4. The mixing effect of coal tar and additives in SKSM

The distribution of coal tar in SKSM is shown in Fig. 10. Since the feed ratio of coal tar and additives is 0.84 to 0.16, the cross section is mainly coal tar phase. The mixing degree of the two fluids before entering the mixing element is better than that of the other three SMs, and the spiral structure may lead to the backflow of the fluids colliding with the pipe wall after passing through the flow channel. The distribution range of coal tar phase volume fraction in SKSM is mainly 52.63% to 89.47%. The distribution trend of coal tar is that the volume fraction at the bottom of the pipeline is the largest and gradually decreases upward. After passing through a single mixing element, the interface gradually transitions downward in an arc, the additives in pure form no longer exist at a distance beyond 112.0 mm at the entrance of the main pipe. The coal tar in pure form no longer exists at a distance beyond 361.8 mm at the entrance of the main pipe. By observing the outlet section of the SM with single mixing element and double mixing element, it can be seen that adding a mixing element to divide the initially separated two fluids into four fluids is obviously helpful for the mixing effect, and the volume fraction distribution area of the outlet section of the double mixing element is more concentrated than that of the single mixing element.

5.5. Hybrid performance analysis

The mixing effect of SVSM, SXSM, SLSM and SKSM can be seen directly from the volume fraction cloud diagram of coal tar phase (Fig. 7-10). Researchers often use the separation strength proposed by Danckwerts [33] to characterize the mixing effect. Its mathematical expressed as the ratio of the actual measured concentration variance of mixed fluids to the variance of complete separation system. In order to accurately analyze the mixing effect of four SMs, the separation strength is used for characterization. The calculation formula is:

$$I = \frac{\frac{1}{m-1}\sum_{i=1}^{m}(X_i - \overline{X})^2}{\overline{X}(1-\overline{X})} \tag{8a}$$

Where X_i is the volume fraction value of coal tar at different positions at different locations of the mixer, and $\overline{X}$ is the average volume fraction of coal tar at m sampling points in a section of the mixer. The separation strength indicates that when the mixing is uniform, the separation strength is 0, and the separation strength is 1 under the condition of complete separation. Therefore, the smaller the separation strength value is, the better is the mixing effect of the fluids. As shown in Fig. 9, for four kinds of static mixers, a section is cut from the inlet of 80 mm at each interval of a mixing element length. The separation strength I of each section is solved by formula (8), and then the variation relationship between the separation strength and the mixing axial position of coal tar and additives flow through different types of SM is drawn from the solved data.

As seen in Fig. 11, before the fluids enter the first mixing element, due to the action of gravity, coal tar and additives have been mixed to a certain extent, and the separation strength of the fluids are lower than 0.74. Before entering the mixing element, due to the special structure of SK and SV mixing elements, the mixing degree is better than SL and SX mixing elements. After the fluids pass through the two mixing elements, it can be seen that the separation strength of SXSM is significantly lower than that of SVSM, SLSM and SKSM. Hence, the mixing effect is better. When the fluids are 160mm away from the coal tar inlet, the separation strength in the four SMs drops below 0.3, and the separation strength of the fluids flowing through SXSM has been as low as 0.009, which is lower than that at the outlet interface of SKSM and SVSM. With the fluids flowing along the axial direction, the separation strength of SLSM at the outlet interface is also reduced to less than 0.01. Therefore, the selection of SXSM saves the length of about 6 mixing elements, which is conducive to the development of coal tar pretreatment equipment with compact structure.

5.6. Influence of the number of SX mixing elements on mixing effect

According to Fig. 7-11, SXSM is the most suitable for mixing coal tar and additives among the four SMs simulated. Therefore, continue to analyze the influence of adding different numbers of SX mixing elements in the SM on mixing effect of coal tar and additives and pressure drop. In addition, the influence of the installation distance between SX mixing

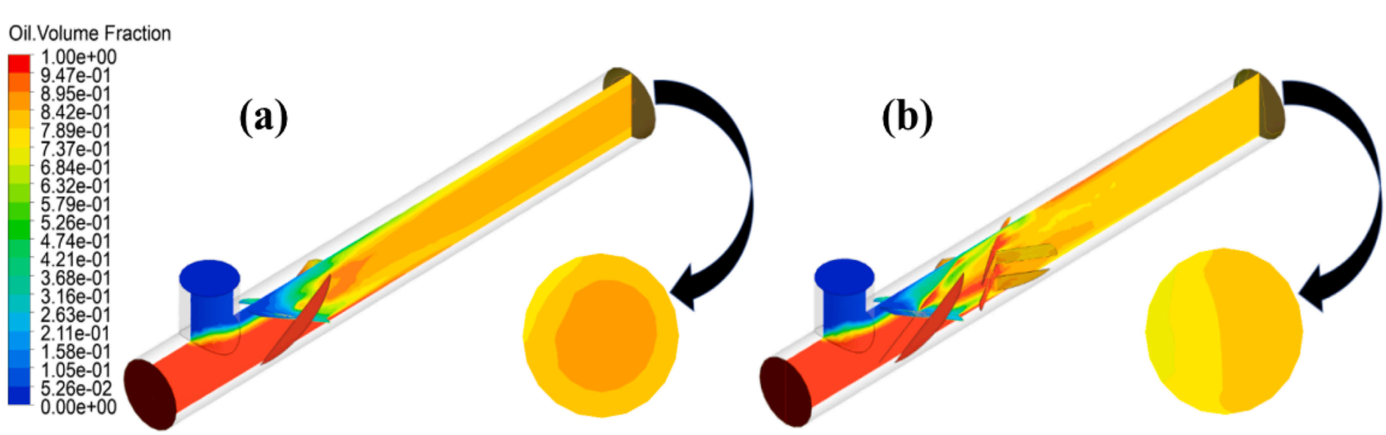

Fig. 9. Volume fraction of coal tar phase in SLSM (a for one SL mixing element, b for two SL mixing elements).

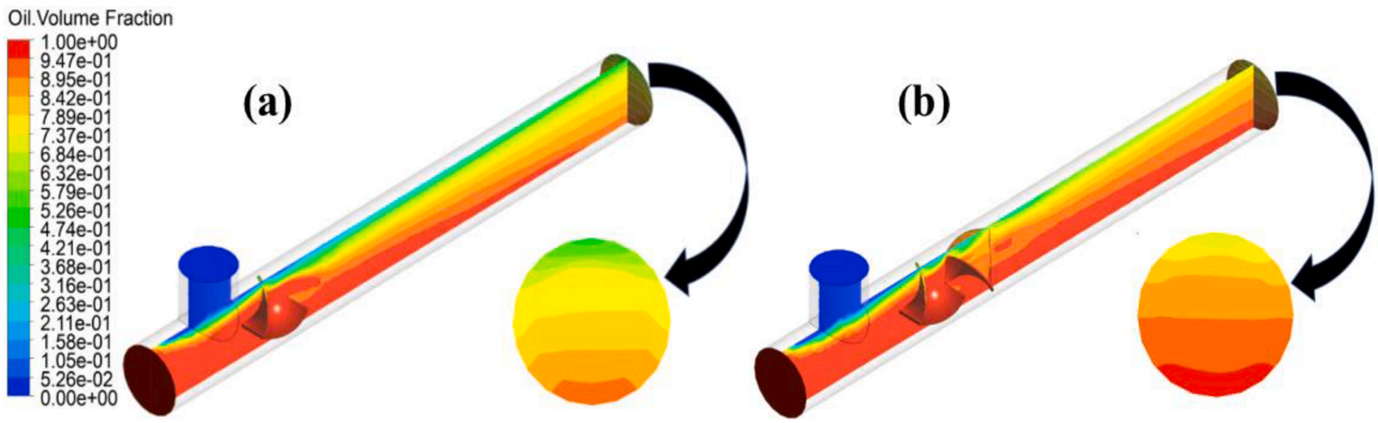

Fig. 10. Volume fraction of coal tar phase in SKSM (a for one SK mixing element, b for two SK mixing elements).

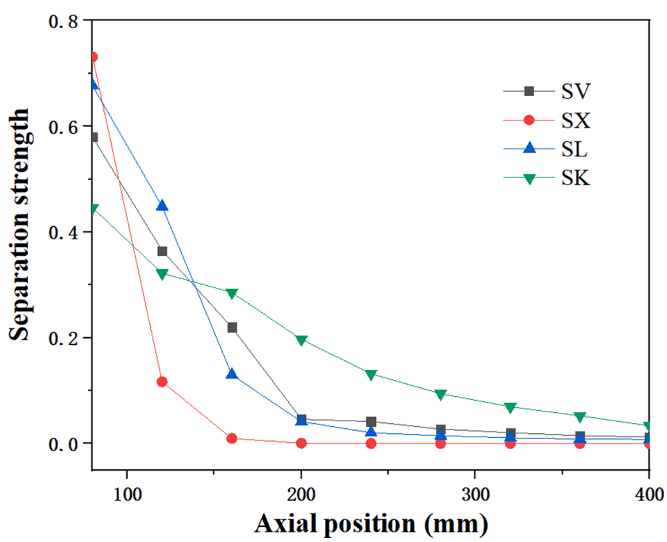

Fig. 11. Variation of separation strength with axial position.

elements on the mixing effect was also investigated.

5.7. Mixing performance

As shown in Fig. 12, when the mixing elements are single or two, although the coal tar and additives have begun to mix, there is coal tar accumulation at the bottom of the pipeline. With the increase of the number of mixing elements, the coal tar accumulation effect at the pipeline edge can disappear faster and faster. This shows that additives has been dispersed into coal tar. From the cross section, the volume fraction of coal tar phase has been basically homogenized in a certain range when the number of SX mixing elements in the pipeline is two, three and four. Therefore, the scale bar interval at the export section is set between 0.82 and 0.85 for observation. As shown in Fig. 13, the increase of mixing elements can increase the homogenization effect. When four mixing elements are installed in the pipeline, the coal tar phase can be stable between 83.7% and 84.5%. This is very close to the previously set coal tar accounting for 84% of the total feed, and the mixing effect is very good.

5.8. Analysis of pressure drop and turbulent kinetic energy

When the installation distance of SX mixing elements is constant, the number of mixing elements can directly affect the pressure drop in the pipeline, and affect the energy consumption. Fig. 14 shows the pressure in the SM shows a downward trend with the increase of the fluids flow distance. Whenever passing through a mixing element, there will be an increase in the pressure drop. This is because SX component affects the flow velocity fluctuation in many directions, and the turbulence of fluids increases, as shown in Fig. 15, so the pressure drop in the pipeline increases. When there is only one mixing element in the pipeline, the pressure loss is 472.35Pa, 932.76Pa for two mixing elements, 1363.22Pa for three mixing elements and 1768.86Pa for four mixing elements.

If only the improvement of mixing efficiency is considered, the more mixing elements installed in the pipeline, the better. However, the more mixing elements, the greater the pipeline pressure loss is. If the pressure loss is too large, it cannot be adapted to the production demand, resulting in waste of energy. Considering the mixing efficiency of a static mixer equipped with three or four mixing elements has little difference, and the pressure drop of static mixers with four mixing elements is 29.7% higher than that of static mixers with three mixing elements. So, it is decided to install three SX mixing elements in the pipeline.

5.9. Influence of installation spacing of mixing elements on mixing effect

In this section, the influence of the direct installation distance of components on the mixing effect is explored by simulating the installation spacing of two SX mixing elements in three groups of pipelines is 0mm, 20mm and 40mm respectively.

As shown in Fig. 16, due to different installation distances, when the mixed fluids flow out of the outlet, its coal tar volume fraction does not depend entirely on gravity and presents a vertical distribution, but will appear to counterclockwise rotation trend. In terms of mixing effect, the effect of coal tar mixing without gap installation is the best, and the effect is the worst among the three groups when the installation distance is 20mm. It is worth noting that when the installation distance continues to extend to 40mm, the mixing effect is improved, but its effect is still a gap compared with gapless installation.

We think that when the installation spacing is 0 mm, the fluids flow through two adjacent mixing elements, and the violent turbulent motion will make the fluids easier to mix. When the installation spacing is 20mm, the fluids pass through the second mixing element gently, and the mixing effect will be worse. When the installation spacing is 40mm, the fluids passe through a mixing element and then stabilizes for a long distance, which may be helpful for mixing, and the effect is slightly better when the fluid passes through the second mixing element.

5.10. Experimental result

Coal tar and additives were pretreated in pilot plants loaded with different SMs. The content of Fe heteroatoms in coal tar before and after reaction is shown in Fig. 17.

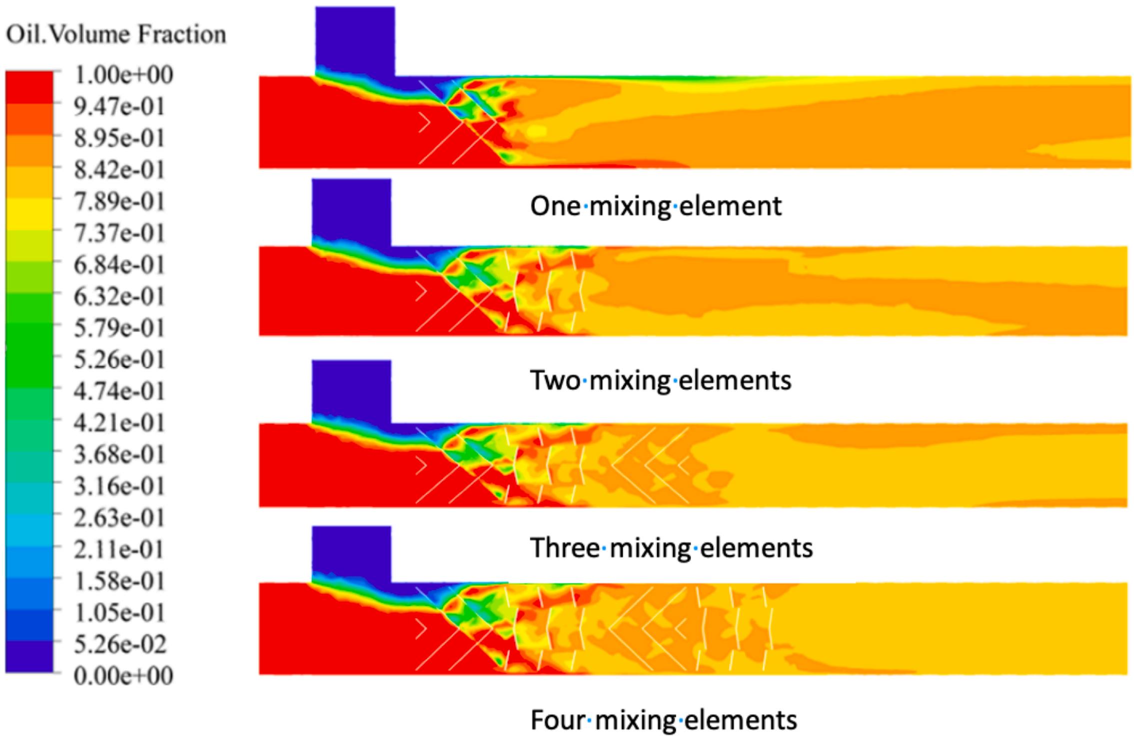

Fig. 12. Volume fraction cloud diagram of coal tar phase on the cross section in SXSM with different number of mixing elements.

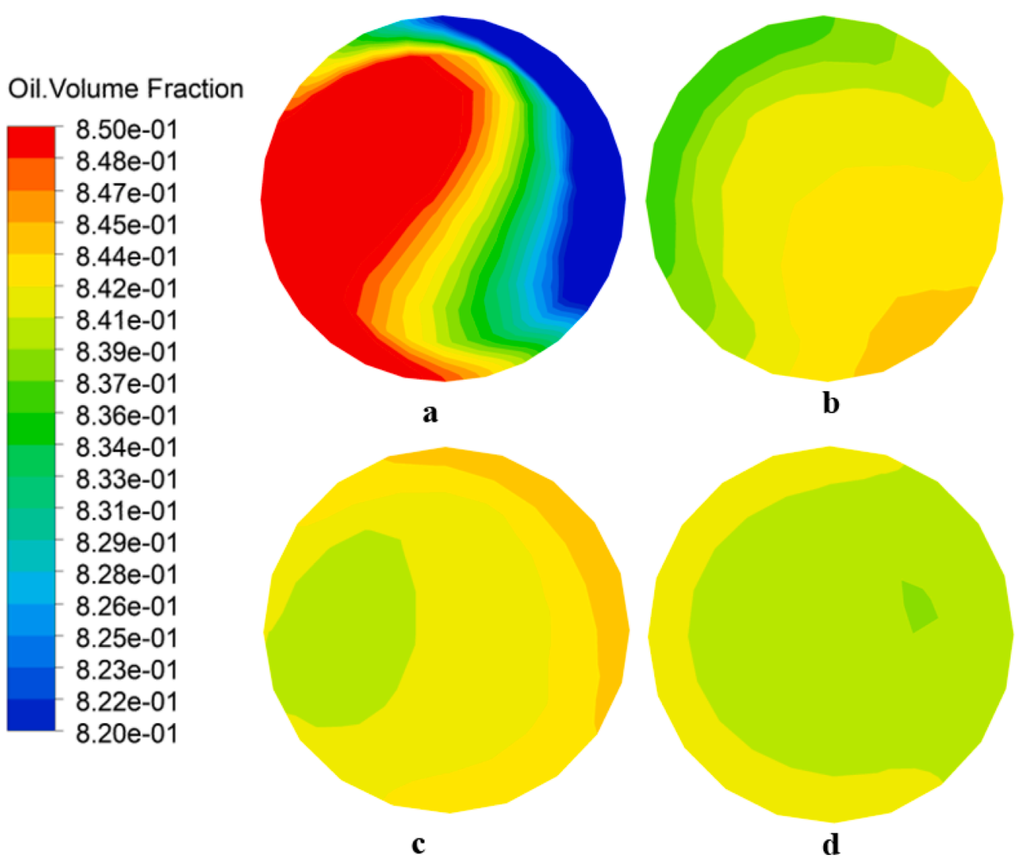

Fig. 13. Volume fraction cloud diagram of coal tar phase on outlet section of SXSM with different number of mixing elements (a for one element, b for two elements, c for three elements and d for four elements).

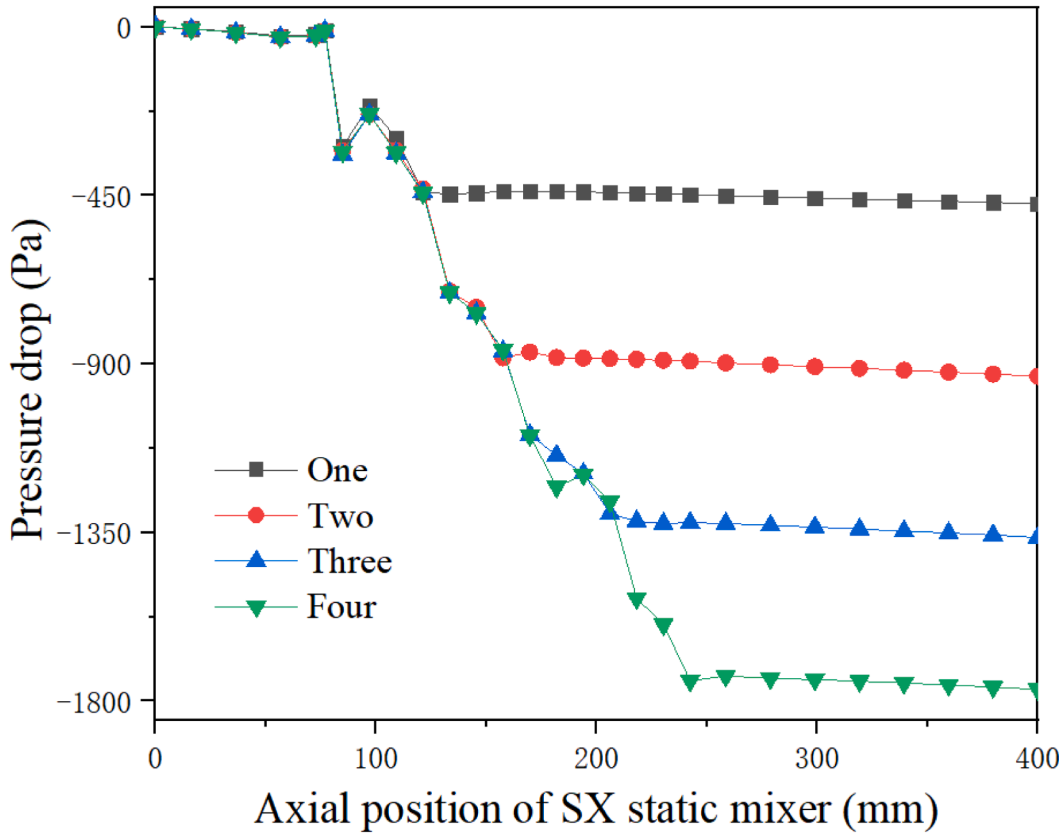

Fig. 14. Axial pressure drop variation of SXSM with different number of mixing elements.

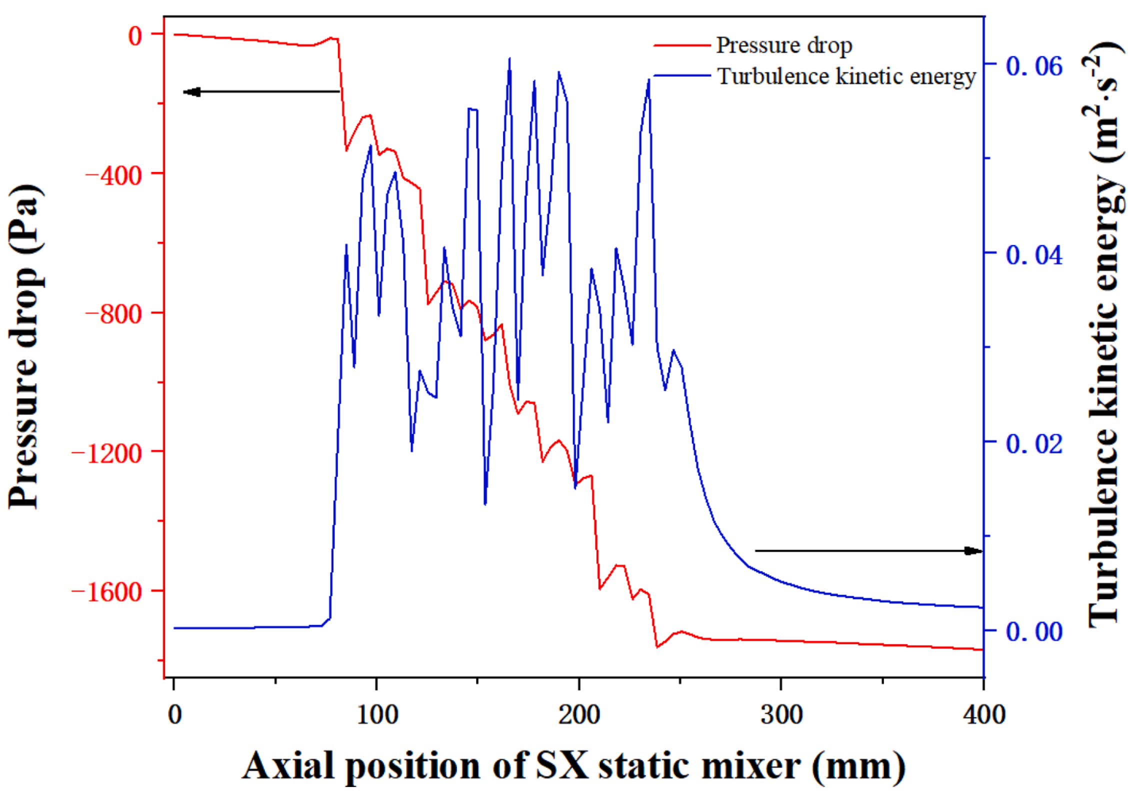

Fig. 15. Variation of turbulent kinetic energy and pressure drop in SXSM.

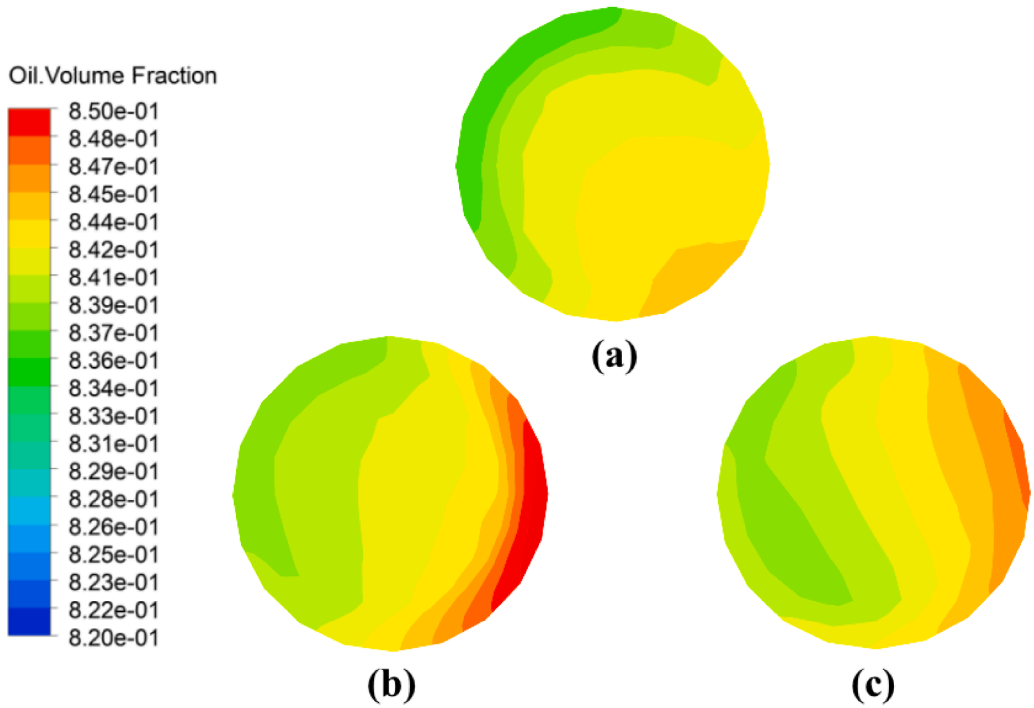

Fig. 16. Volume fraction of coal tar phase in outlet section of SXSM loaded with mixing elements with different installation spacing (a is no gap, b is the spacing of 20mm, c is the spacing of 40mm).

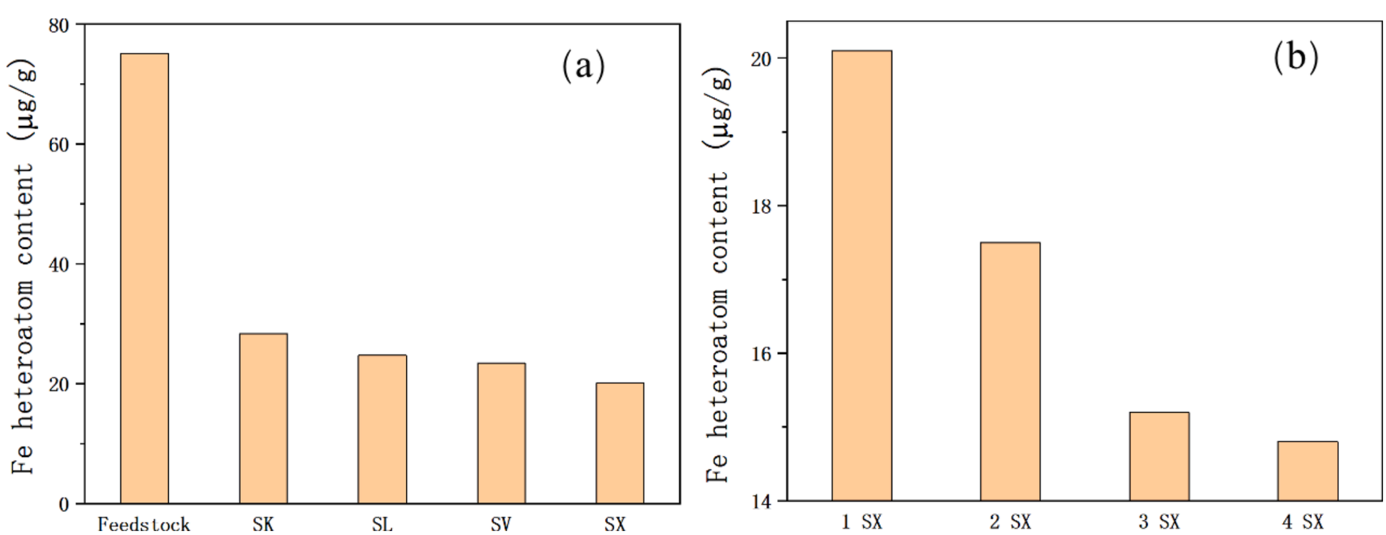

Fig. 17. The Fe heteroatom content of coal tar before and after pretreatment in pilot plant equipped with different SMs (Figure a shows that the pretreatment pilot plant is loaded with SVSM, SXSM, SLSM and SKSM containing one mixing element, Figure b shows the pretreatment pilot plant loading SXSM with one, two, three and four mixing elements).

5.11. Comparison between the simulation and experimental results

The pilot plant equipped with SVSM, SXSM, SLSM and SKSM is selected for confirmatory comparison test. In order to ensure the parallel comparison between test results and simulation results, other process conditions remain unchanged. The test results are shown in Fig.17 (a). After pretreatment in the pilot plant loaded with SK, SL, SV and SXSMs, the Fe heteroatom content of coal tar decreases in turn. Although most of the heteroatoms are removed from the raw materials, the pretreatment effect of SXSM is the best under the same number of mixing elements. Compared with the volume fraction cloud diagram of coal tar phase, it can be seen that the experimental results and numerical simulation results show that SXSM has the best mixing effect. In order to further verify the simulation results of the number of mixing elements, confirmatory comparative tests are carried out on the pilot plant loaded with SXSM containing one, two, three and four mixing elements, and other process conditions remain unchanged to ensure the parallel comparison between the test results and the simulation results. The test results are shown in Fig.17 (b). The Fe heteroatom content in coal tar decreases with the increase of the number of mixing elements. Because the separation effect remains unchanged, it shows that the mixing effect is better with the increase of the number of mixed elements. And the heteroatom removal effect, that is, the pretreatment effect is better. When the fourth mixing element is added, the Fe heteroatom content decreases slightly. The simulation results also show that the mixing effect is better when the number of SX mixing elements is 3 or 4.

6. Conclusion

Three dimensional CFD simulation of incompressible turbulent fluids in SM, the key equipment of pilot scale-up experiment of coal tar pretreatment, was carried out by using the CFD software. By analyzing the volume fraction and pressure loss of coal tar-additive two-phase system, the suitable type, number of internals and installation spacing of SM are obtained. The main conclusions are as follows:

(1) SV, SX, SL or SK mixing elements can realize the gradual separation, rotation and convergence of fluids, and change the velocity and direction of fluids at each position, so as to achieve the purpose of effective and uniform mixing. SKSM and SVSM can cut the fluids, but the guided fluids disturbance mainly has only one axial dimension, and the guided flow amplitude in other dimensions is small. SXSM and SLSM can use complex structures to cut the fluids in the pipeline and guide the fluids to flow in axial and longitudinal dimensions. When installing double mixing elements, the installation in 90° opposition can effectively solve the problem of single flow disturbance direction and make the flow more stable.
(2) From the volume fraction cloud diagram of coal tar phase, the flow of coal tar and additives in four SMs can be observed intuitively. For SVSM, SKSM, SLSM and SXSM, the axial positions at which pure additives no longer exist are 146.6mm, 112.0mm, 113.8mm and 96.3mm respectively, and the axial positions where pure MTCT no longer exist are 400mm, 361.8mm, 207.1mm and 190.7mm respectively. It was found that the mixing effect of SX mixer is better.
(3) Then the separation strength is used to analyze the mixing effect of four SMs. It is found that the minimum separation strength of the outlet section of SXSM is 3.5×10^{-5}, the mixing uniformity is the best, the efficiency is the highest and the stability is the strongest. This is because the SX type mixing element has a complex geometric structure of "multi-X" type, which guides the fluids flow towards axial and longitudinal dimensions after cutting. It is helpful for the full mixing of coal tar and additives, which are medium and high viscosity liquid medium, and the subsequent pretreatment effect can be improved, so that the pretreated coal tar can meet the requirements of subsequent deep processing and utilization.
(4) Through further analysis and discussion, several groups of SXSM with different number of mixing elements and different installation distance are simulated by CFD. By analyzing the volume fraction and pressure drop, it revealed that when the number of mixing elements is 3 and the installation rotation is 90 ° without gap, the comprehensive performance is better.
(5) Finally, through a verification experiment, the removal effect of Fe heteroatom after pretreatment was regarded as the pretreatment effect as the detection index to compare the mixing and reaction effect of coal tar and additives in different SMs. The verification experiment also shows that SXSM is more suitable for the fully mixing medium and high viscosity liquid medium such as coal tar and additives. When the number of mixing elements is 3, the comprehensive benefit is the best, which is consistent with the numerical simulation results.

CRediT authorship contribution statement

Yijie Wang: Writing – review & editing, Methodology, Investigation, Writing – original draft. **Dong Li:** Project administration, Investigation, Resources. **Yonghong Zhu:** Software, Validation. **Jieping Liu:** Validation. **Hongyu Chen:** Software. **Qing Guo:** Data curation. **Chongpeng Du:** Validation. **Zhengpeng Miao:** Formal analysis. **Louwei Cui:** Supervision, Resources. **Feng Tian:** Software. **Jie Liu:** Software. **Huaan Zheng:** Project administration.

Declaration of Competing Interest

The authors declare no conflict of interest.

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